

Quiz 2 — Practice Problems

Reference: Banda-Barragán, W. E. (2026). *Computational Physics 1: Unit 2 lecture notes*. GitHub. <https://github.com/wbandabarragan/computational-physics-1/tree/main/unit-2>

How to work through it

- **The quiz will be written, and taken without a computer, a notebook or a phone.** You will answer concept questions, hand-write pseudocode, and sketch workflow diagrams.
- **Practise writing pseudocode by hand.** Numbered steps, one action per line, the loops and the conditionals written out explicitly, and every function named with inputs and outputs. The logic has to be complete.
- Run each notebook of Unit 2, then understand and replicate the code. Practise coding!
- Submission of these problems is not needed. This sheet is practice for Quiz 2.

Part A — What to study

- **201 — Python basics.** Data types, operators, variables, and the four data structures: lists, tuples, sets and dictionaries. What each one is for, and which are mutable.
- **202 — NumPy arrays.** 1-D and 2-D arrays, array generation, indexing and slicing, vectorised operations, boolean indexing as a conditional. Why use vectorisation.
- **203 — Plotting.** The figure environment, overplotting, subplots, scatter plots, histograms and their bin midpoints, axis labels, legends and saving a figure.
- **204 — Functions.** Built-in, third-party and your own; arguments and multiple returned objects; lambda, nested and recursive functions; a function passed as an argument .
- **205 — Data I/O.** Number formatting, the `open` function and its modes, reading and writing text and CSV files, writing a table of numbers.
- **206 — Data I/O with pandas.** Series and DataFrames, labels and named indexes, locating rows, reading a file into a DataFrame, building one and writing it out.
- **207 — Data fitting and regression.** Correlation and monotonicity (Spearman, Pearson), the LM method, linear regression with Numpy and SciPy, regression uncertainties.
- **208 — χ^2 fitting.** The definition of χ^2 , functionals and optimisation, the reduced χ^2 and what its value tells you.
- **209 — Interpolation.** Global interpolation (Newton, Lagrange) versus point-to-point interpolation (linear, quadratic, spline); missing data and NaN handling.
- **210 — Extrapolation.** What changes when you leave the range of the data, and what makes an extrapolation defensible.

Part B — Pseudocode and Workflow Diagrams

Write pseudocode for each of the problems below, and beside it a workflow diagram using the symbols of Lecture 107. Mark the data I/O, the operations, the decisions and the loops, and name every function with its inputs and its outputs.

1. **χ^2 minimisation.** You have N triples (x_i, y_i, σ_i) and a model $f(x; a, b)$. Write the pseudocode of a program that defines $\chi^2(a, b)$, searches for the pair (a, b) that minimises it, and reports the reduced χ^2 . Say explicitly how your search decides that it has finished.

2. **Missing data and interpolation.** A column of measurements has gaps recorded as NaN. Write the pseudocode of a program that locates the gaps, builds an interpolator from the valid points only, fills the gaps, and then plots the original and the filled data on the same axes so that the filled points can be told apart.